

Omar Jamal

Full-Stack Software Engineer | Cloud, DevOps & Platform Engineering

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PROFESSIONAL SUMMARY

Full-stack software engineer with 3 years of experience building real-time systems, cloud-native backends, and IoT platforms across the full software development lifecycle. Designed and shipped a complete production platform as sole engineer, from an embedded hardware control agent and containerized microservices backend to a voice AI pipeline for a mobile application. Proficient across the entire stack, from low-level Linux systems and embedded hardware to cloud infrastructure, DevOps automation, and AI-integrated application development.

WORK EXPERIENCE

IoT Engineering Intern

December 2025 – May 2026

Dropolabs

Ben Guerir, Morocco

- Delivered a production-ready IoT platform for qPCR machines as sole engineer by owning the complete stack, from an edge device agent for direct hardware control to a containerized cloud backend with live telemetry streaming
- Secured device-to-cloud communication by implementing JWT authentication with an auto-registration mechanism, eliminating manual hardware provisioning
- Enabled reliable experiment lifecycle management by designing a FastAPI and PostgreSQL backend with 20+ REST API endpoints and Docker multi-service orchestration
- Reduced release friction by establishing GitHub Actions CI/CD pipelines, automating end-to-end testing and deployment across all platform services

AI Engineering Intern

May 2025 – October 2025

SMValley

Agadir, Morocco

- Enabled hands-free application navigation for elderly users by building a voice pipeline connecting speech-to-text (STT), a GPT-based conversational agent, and text-to-speech (TTS) over real-time LiveKit WebRTC audio streams
- Accelerated feature extensibility by designing a Python decorator-based tool-calling framework, enabling the agent to dynamically invoke healthcare actions including appointment scheduling and medication reminders
- Improved platform accessibility by integrating Darija and French multilingual support across speech recognition and large language model (LLM) prompt layers for a React Native mobile application

PROJECTS

FT_SERVICES – Kubernetes Microservices Cluster | *Docker, Kubernetes*

Designed a cloud-native microservices cluster and automated rolling deployments with horizontal scaling across all services, achieving near-zero downtime.

Pongify – Real-Time Multiplayer Game | *WebSockets, Redis, PostgreSQL*

Built a real-time multiplayer system and reduced network latency by 20% through optimized WebSocket connection management and Redis-backed session handling.

FT_IRC – IRC Server in C++ | *C++, TCP/IP*

Engineered a concurrent IRC server supporting 50+ simultaneous users via non-blocking I/O and efficient socket multiplexing.

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, Bash

Frameworks: FastAPI, Django, React, React Native

Databases: PostgreSQL, Redis, MongoDB

DevOps & Cloud: Docker, Kubernetes, CI/CD, GitHub Actions, Cloud-Native Development, Microservices

Engineering: REST API Design, WebSockets, Real-Time Systems, IoT, Platform Engineering, Embedded Systems, System Design, Linux, Git

Soft Skills: Cross-Functional Collaboration, Problem-Solving, Adaptability, Technical Communication, Initiative

EDUCATION

1337 Coding School (42 Network)

Computer Architecture & Systems Engineering

Ben Guerir, Morocco

October 2022 – Present

LANGUAGES

English (Professional) | Arabic (Native) | French (Basic)